

# A Wonder is not built. It is noticed.

There is no “build a Wonder” button. You dig, you raise, you hang a cloud, you sow. The water works out where to go. At the Harvest the game looks at the board and says: **three hexes of still water, side by side — that is a Lake.**

You never made a lake. You made a hollow, and it rained. The Wonder is the **name the game gives to a shape the world arrived at**. That is the whole idea, and everything below follows from it.

## Where it sits in the score

Bloom

=

worth of living hexes  
what you grew

×

( 1 + the Wonders standing )  
what the water made of it

)

**Life is the count. Wonders are the multiplier.** They are kept apart on purpose. If they were one number, “cover the whole pot in moss” would be the best play — one move, repeated, forever. Split in two, you have to do both: grow something **and** leave the land interesting. Wonders are almost all about water and terrain, so they cannot be farmed by sowing.

## Two real readings, measured in the build

seven living hexes that happen to touch – a Grove

$7 \times ( 1 + 0.25 ) = \text{Bloom } 8.8$

the same twelve hexes around a lake – Lake + Ring of Life + Grove

$12 \times ( 1 + 0.5 + 1.0 + 0.25 ) = 12 \times 2.75 = \text{Bloom } 33$

Same amount of green. Nearly four times the score, because of the shape it grew into.

**It is a pure read**

Nothing is stored. wonders() looks at the board as it stands and returns a tally. Build the same terrain twice and you get the same Wonders. It runs at the Harvest and when asked — 14 ms on the largest board.

**It can be lost**

Drain the lake and the Lake is gone. What stands at the Harvest is what counts; there is no “I made one once, it is banked”. This falls out of it being a read rather than a record.

**The Codex is a field guide**

Not an inventory. Gold means found before in this browser; green means standing in the pot right now. Each entry carries how it is made, so it is a list of things to aim at.

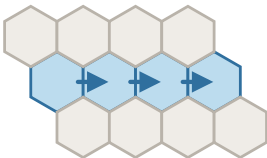
## What they look like

**The Lake**



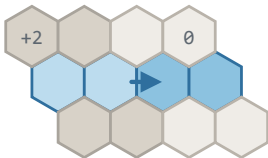
Three hexes of still water, touching.

**The River**



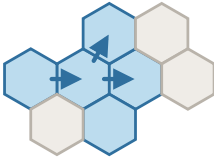
Four hexes in a row that water runs through.

**The Waterfall**



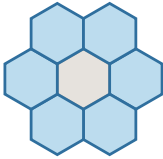
A river pouring onto ground two steps below.

**The Fork**



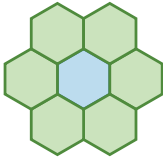
One river hex sending water out two ways.

**The Island**



Dry ground ringed all the way round by water.

**The Ring of Life**



A lake with life or water on every hex around it.

■ water

■ life

■ stone

■ dry ground

— arrows are water leaving through an edge, which the board already stores as out[6].

## The twelve, and what each is worth

Every line is a restatement of the detector in the code, and has to stay true to it. The four marked **+** **stack**: three waterfalls count three times. The other eight count once however many you have.

| WONDER                 | HOW IT IS MADE                                                           | BONUS |
|------------------------|--------------------------------------------------------------------------|-------|
| The Lake               | Three hexes of still water side by side.                                 | +0.5  |
| The Great Lake         | A lake of twelve hexes or more. (Replaces the Lake, not added to it.)    | +1.0  |
| The River              | Four hexes in a row that water runs through.                             | +0.5  |
| The Waterfall <b>+</b> | A river pouring onto ground two steps below.                             | +0.5  |
| The Fork <b>+</b>      | One river hex sending water out two ways.                                | +0.25 |
| The Delta <b>+</b>     | A fork within two hexes of two separate ways off the board.              | +0.75 |
| The Island             | Dry ground ringed all the way round by water.                            | +1.0  |
| The Ring of Life       | A lake with life or water on every hex around it.                        | +1.0  |
| The Grove <b>+</b>     | Seven living hexes joined together.                                      | +0.25 |
| The Terrace            | Five living hexes, each with higher ground one side and lower the other. | +0.5  |
| The Highland Meadow    | Three living hexes at three steps up or higher.                          | +0.5  |
| The Watershed          | One cloud whose water reaches two far-apart edges of the board.          | +0.75 |

### Why these twelve

- **Every one is already computable.** No Wonder needed a new quantity in the simulation. They read `pool`, `out[6]`, `ground height` and `life` — all of which the board keeps anyway.
- **They span the verbs.** Lake and River reward water routing; Waterfall and Terrace reward relief; Grove and Highland Meadow reward sowing; Delta and Watershed reward reading the whole board at once.
- **They disagree with each other.** A Lake wants water to stop; a River wants it to move; an Island wants a hole in the lake. You cannot get all twelve on one pot, so a run has a shape.
- **The hard ones arrive late**, which is what makes the Omen and the Whim spread themselves over the eight Years rather than firing in Year 1.

### What they are not

- **Not objects.** Nothing is placed. There is no Lake entity anywhere in the code — only water that happens to be still, in a group of three.
- **Not achievements.** They score every Year, every time, and they can go away.
- **Not random.** Nothing about a Wonder is rolled. The same board gives the same answer, always.
- **Not the goal.** The bar is Bloom. Wonders are how a small amount of life becomes a large amount of Bloom — they are leverage, not the target.

### Two places they show up

- **The Omen** — each Year, Gaia asks for one of them, you choose which, and the Harvest pays dew.
- **Gaia’s Whim** — every Wonder turns the wheel the first time it stands in a run.

Detectors: `wonders()` in `index.html` • score: `harvest()` • recipes: `WONDER_HOW` • bonuses: `BONUS`. The Codex screen in the game shows this same table, live.